

- 1 (a) Draw a ring around the irrational number in this list.

$$\frac{4}{5}$$

$$0.12$$

$$2\pi$$

$$-4.5$$

[1]

- (b) Calculate $\sqrt{2} \times \sqrt{32}$

..... [1]

- 2 Write 30 000 in standard form.

..... [1]

- 3 Chen tries to factorise four expressions.

Tick (✓) the **two** expressions that he has factorised correctly.

Expression	$3x^2 + 6x$	$8a^2 + 24a$	$6y^2 + 9y$	$14b^2 - 21b$
Answer	$3x(x + 2)$	$4a(2a + 8)$	$6y(y + 3)$	$7b(2b - 3)$
	☐	☐	☐	☐

[1]

- 4 Tick (✓) to show if each of these statements is true or false.

	True	False
All squares are also rectangles.	☐	☐
All parallelograms are also rhombuses.	☐	☐

[1]

- 5 Write a number in the box to make this statement correct.

$$8^{\boxed{}} = \frac{1}{8^4}$$

[1]

- 6 Insert **one** pair of brackets to make this statement true.

$$14 + \sqrt{36} \div 2 = 10$$

[1]

- 7 Triangle A is reflected in the line $x = -3$ to give triangle B.

Samira says,

‘Triangle B must be reflected in the line $y = -3$ to give triangle A.’

Samira is **not** correct.

Complete Samira’s statement with the correct line of reflection.

Triangle B is reflected in the line to give triangle A.

[1]

- 8 Mike thinks of a number, n .
He adds 3 to his number.
He then squares the result.

Write an expression, in terms of n , to represent the number Mike finishes with.

..... [1]

9 Write a number in each box to complete the statements.

$$(k + 5)(k + 9) = k^2 + \boxed{}k + 45$$

$$(p + 6)^2 = p^2 + 12p + \boxed{}$$

[2]

10 Write a **whole number** in the box to complete the statement.

$$4 < \sqrt{\boxed{}} < 4.5$$

[1]

11 (a) The n th term of a sequence is $\frac{n}{2}$

Find the value of the 12th term of the sequence.

..... [1]

(b) Here are the first five terms of a different sequence.

1, 8, 27, 64, 125

Find the n th term of this sequence.

..... [1]

- 12** The scale drawing shows the position of a lighthouse, L .
The scale is 1 cm represents 10 km.



Scale: 1 cm = 10 km

A boat is 85 km from the lighthouse on a bearing of 110° .

On the scale drawing, mark the position of the boat with a cross ($\times$). [2]

- 13 (a)** Find the size of an exterior angle of a regular pentagon.

..... $^\circ$ [1]

- (b)** A regular polygon has an interior angle of 170° .

Work out the number of sides of the polygon.

..... [2]

- 14** On Monday Lily records the colours of cars passing her window. She writes the relative frequencies as probabilities in the table.

Colour of car	Red	Blue	Grey	Other
Probability		0.2	0.18	0.45

- (a)** Complete the table.

[1]

- (b)** On Tuesday 600 cars pass Lily's window.

Calculate the expected number of cars passing Lily's window on Tuesday that are blue or grey.

..... [2]

- 15** Tick (✓) to show if each of these polygons is regular or not regular.

Number of sides of polygon	Symmetry properties of polygon	Regular	Not Regular
4	2 lines of symmetry and rotational symmetry order 2		
4	4 lines of symmetry and rotational symmetry order 4		
10	10 lines of symmetry and rotational symmetry order 5		

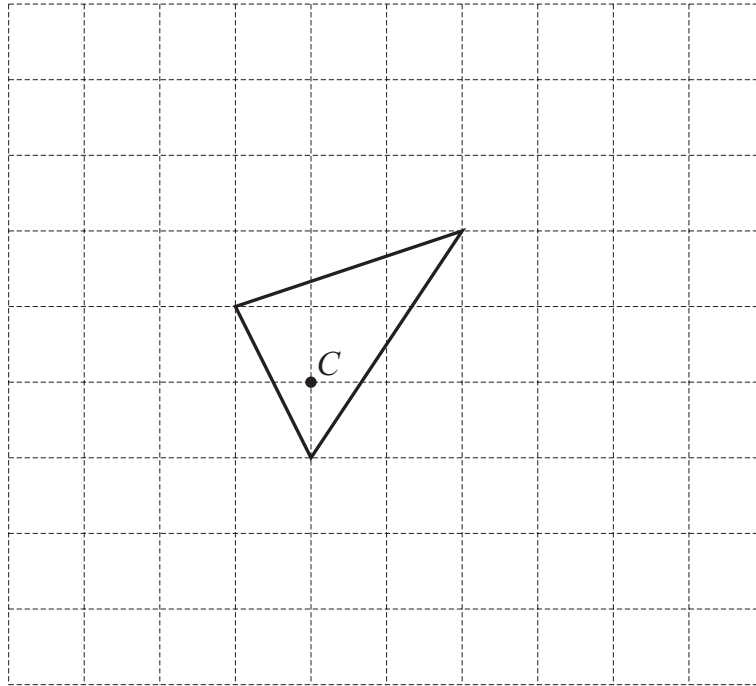
[1]

- 16** The length, L metres, of a car is 3.2 metres correct to one decimal place.

Complete the inequality to show the bounds of L .

$$\dots\dots\dots \leq L < \dots\dots\dots \quad [2]$$

- 17** A triangle is drawn on the grid.



On the grid, draw the enlargement of the triangle by scale factor 2 from centre C . [2]

18 Solve these simultaneous equations.

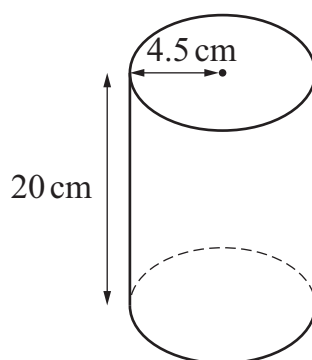
$$\begin{aligned} 3x + 8y &= 28 \\ x + 4y &= 16 \end{aligned}$$

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

[3]

19 The diagram shows a cylinder with a radius of 4.5 cm and a height of 20 cm.



NOT TO
SCALE

Calculate the volume of the cylinder.

..... cm³ [2]

20 Rearrange the formula $y = 6x^2 - a$ to make x the subject.

$x =$ [3]

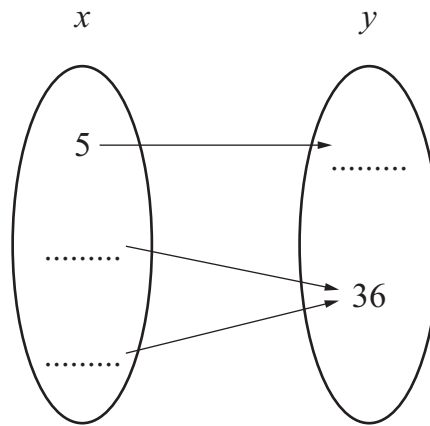
21 Jamila attempts to round the two numbers in the table correct to 3 significant figures. Her answers are wrong.

Complete the table with the two numbers correctly rounded to 3 significant figures.

Number	Jamila's answer	Number correct to 3 significant figures
207 499	207 500	
0.03478	0.03	

[2]

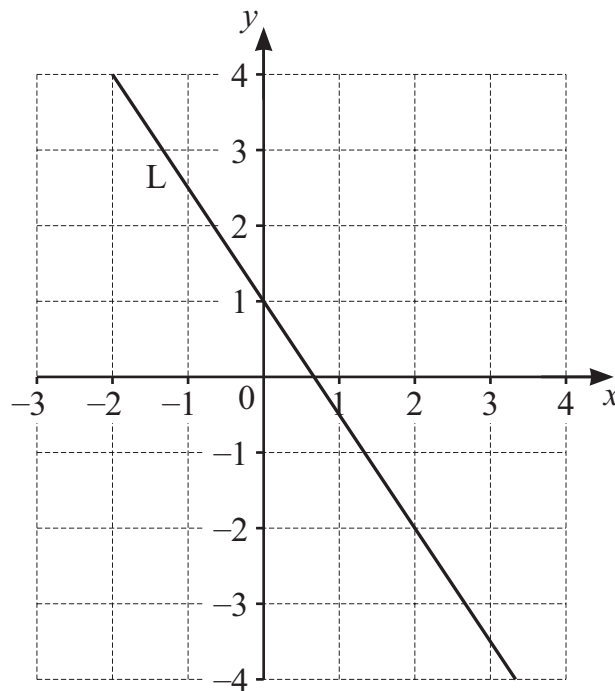
22 This mapping diagram is for the function $y = 4x^2$



Complete the diagram with **three** different values.

[3]

23 Line L is drawn on the grid.

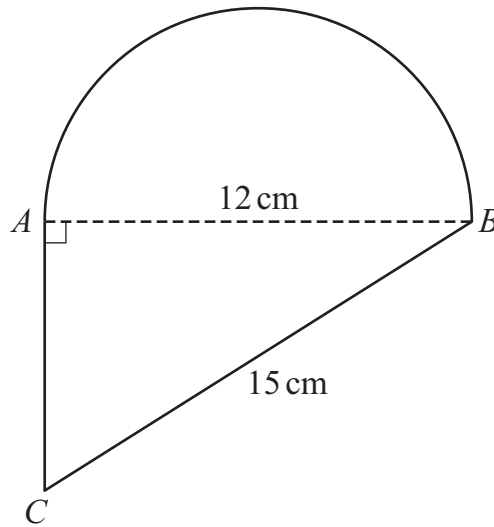


Find the equation of line L.

Give your answer in the form $y = mx + c$

$y =$ [2]

- 24 The diagram shows a shape formed by a semicircle with diameter AB joined to a right-angled triangle ABC .



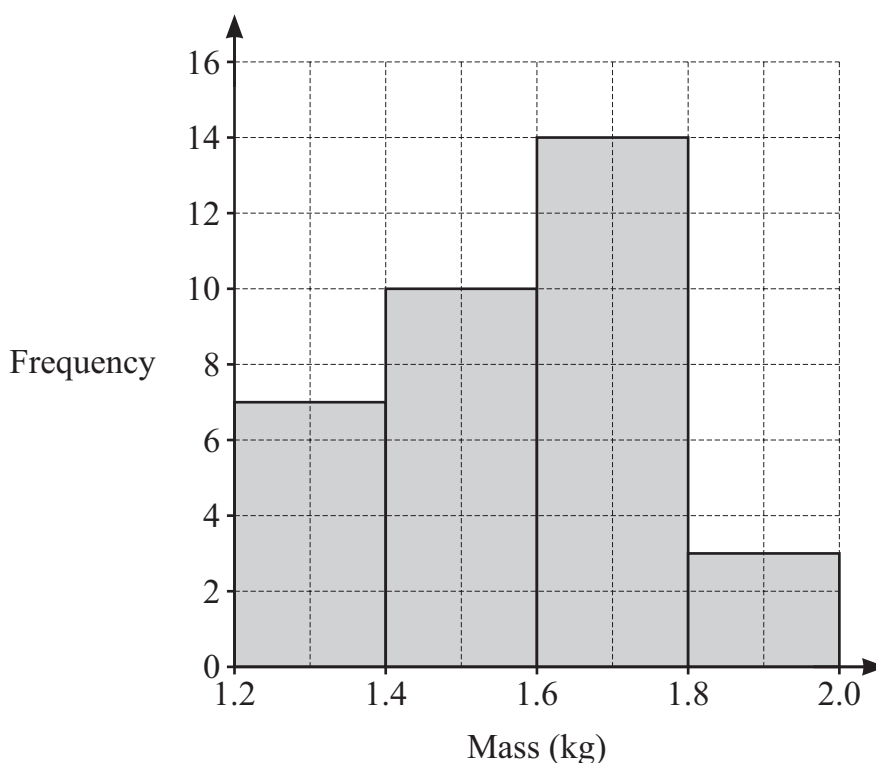
NOT TO
SCALE

$AB = 12$ cm and $BC = 15$ cm.

Calculate the total area of the shape.

..... cm^2 [4]

25 The graph shows information about the mass of each of 34 melons.



Calculate an estimate of the mean mass of these melons.

..... kg [4]

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